

SMART FIRE AND GAS LEAKAGE MONITORING SYSTEM USING ESP32

Abstract

This project presents a Smart Fire and Gas Leakage Monitoring System using an ESP32 microcontroller. The system monitors temperature, humidity, smoke, gas leakage, and fire using DHT11, MQ-2, and flame sensors. When hazardous conditions are detected, the ESP32 activates a buzzer, displays alerts on a 16×2 LCD, and controls an exhaust fan through a relay module. The project provides a low-cost, reliable, and real-time safety solution suitable for homes, industries, laboratories, and smart buildings.

Key Features

- Real-time gas and flame detection
- Temperature and humidity monitoring
- Automatic buzzer and LCD alerts
- Relay-controlled exhaust fan
- Low-cost ESP32-based design

Technologies Used

ESP32, Arduino IDE, Embedded C, DHT11 Sensor, MQ-2 Gas Sensor, Flame Sensor, Relay Module, 16×2 LCD Display, Buzzer, DC Fan.

Applications

Home safety, industrial monitoring, laboratories, kitchens, warehouses, and educational IoT projects.

Conclusion

The project demonstrates an efficient embedded safety monitoring system capable of detecting fire and gas leakage at an early stage. It can be extended in the future with IoT cloud monitoring and mobile notifications.